

BURNS

Calculate the percentage total body surface area (% TBSA) burnt using the figure below.

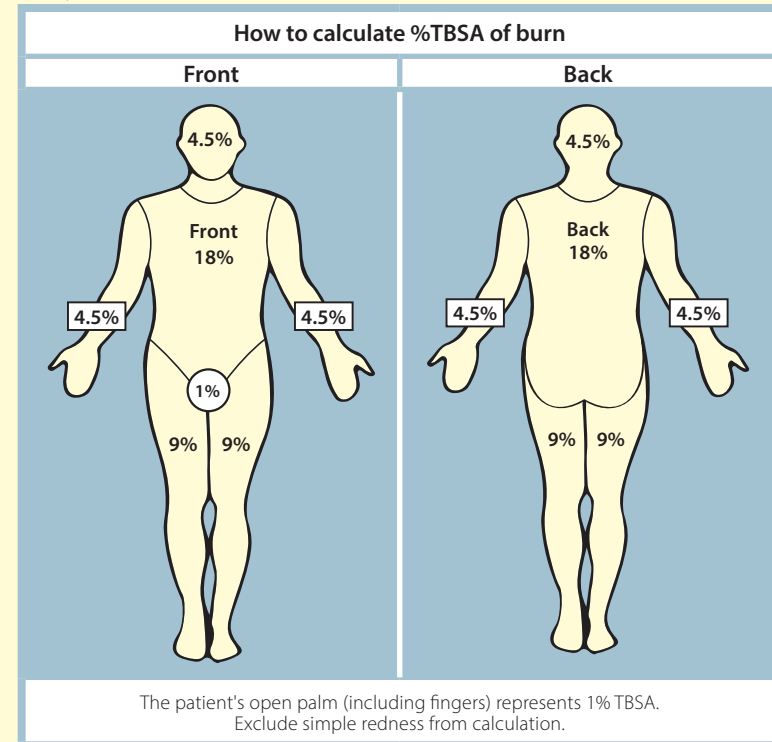
- Drowsy or confused
- Electric/chemical burn
- Full-thickness burn (white/black, painless, leathery, dry)
- Partial thickness burn (pink/red, blisters, painful, wet) > 10% TBSA
- **Inhalation injury** likely (burns to face/neck, difficulty breathing, hoarse, stridor or black sputum)

Management:

- Remove clothing. Cool burn with cool tap water or wet towel/s for 30 minutes. Keep warm with clean, dry sheet.
- Give face mask oxygen if burn > 10% TBSA, inhalation injury, oxygen saturation < 94% or drowsy/confused. Doctor to consider intubation.
- If > 10% TBSA:
 - Insert a large-bore IV line. If % TBSA burnt > 40% or if transport to hospital likely to take more than 45 minutes, insert a second IV line.
 - Give **sodium chloride 0.9%** IV 4mL x weight (kg) x % TBSA over 24 hours. Give half this volume in first 8 hours from time of burn. Calculate the hourly volume (mL) = total volume (mL) ÷ 2 ÷ 8.
 - Insert a urine catheter and document urine output every hour.
- Give **paracetamol** 1g orally 4-6 hourly (up to 4g in 24 hours).
- If pain severe, give **morphine** 3-10mg slow IV¹.
- If other injuries, manage ➤ 18.
- Clean and dress burn gently:
 - Remove loose/dead skin and clean burn with **sodium chloride 0.9%**.
 - If full thickness or > 10% TBSA burn, apply paraffin gauze and cover with plastic wrap.
 - If hospital transfer delayed > 12 hours, apply paraffin gauze and cover with dry gauze and bandage.
 - If none of above, apply Burnshield[®] and cover with bandage. If not available, use a non-adherent dressing or wrap in clean, dry sheet and blanket.
- Give **tetanus toxoid** 0.5mL IM if none in past 5 years.
- Monitor hourly while awaiting transport: BP, pulse, respiratory rate, oxygen saturation, level of consciousness and urine output.
- Refer urgently.

Give urgent attention to the patient with burn/s and any of:

- Circumferential burn of chest/limbs
- Burn to face, hand/foot, genitals, joint
- Oxygen saturation < 94%
- Temperature ≥ 38°C
- BP < 90/60
- Other injury



Approach to the patient with burn/s not needing urgent attention

- Cool burn < 3 hours old with cool tap water or wet towel/s for 30 minutes.
- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- Remove loose/dead skin and gently clean burn with **sodium chloride 0.9%**. Then cover with paraffin gauze dressing.
- Give **tetanus toxoid** 0.5mL IM if none in past 5 years.
- If cigarette burns, burn with specific shape of object (e.g. iron, grid, knife/fork, car cigarette lighter, light bulb), repeated/unexplained burns or other unexplained injuries, consider abuse ➤ 88 and self-harm ➤ 83.
- Review daily until burn healed:
 - Dress burn with paraffin gauze dressing. If signs of infection (redness, swelling), apply **povidone iodine 5%** cream daily.
 - If severe infection (extensive redness or swelling, foul-smell, pus or temperature ≥ 38°C), pain despite medication or burn not healed within 2 weeks, refer

¹Dilute 10mg morphine with 9mL of sodium chloride 0.9%. Give diluted **morphine** 3mL IV over 3 minutes (1mL/minute). If needed, give another 1mL/min until pain improved, up to 10mL. Stop if BP drops < 90/60.